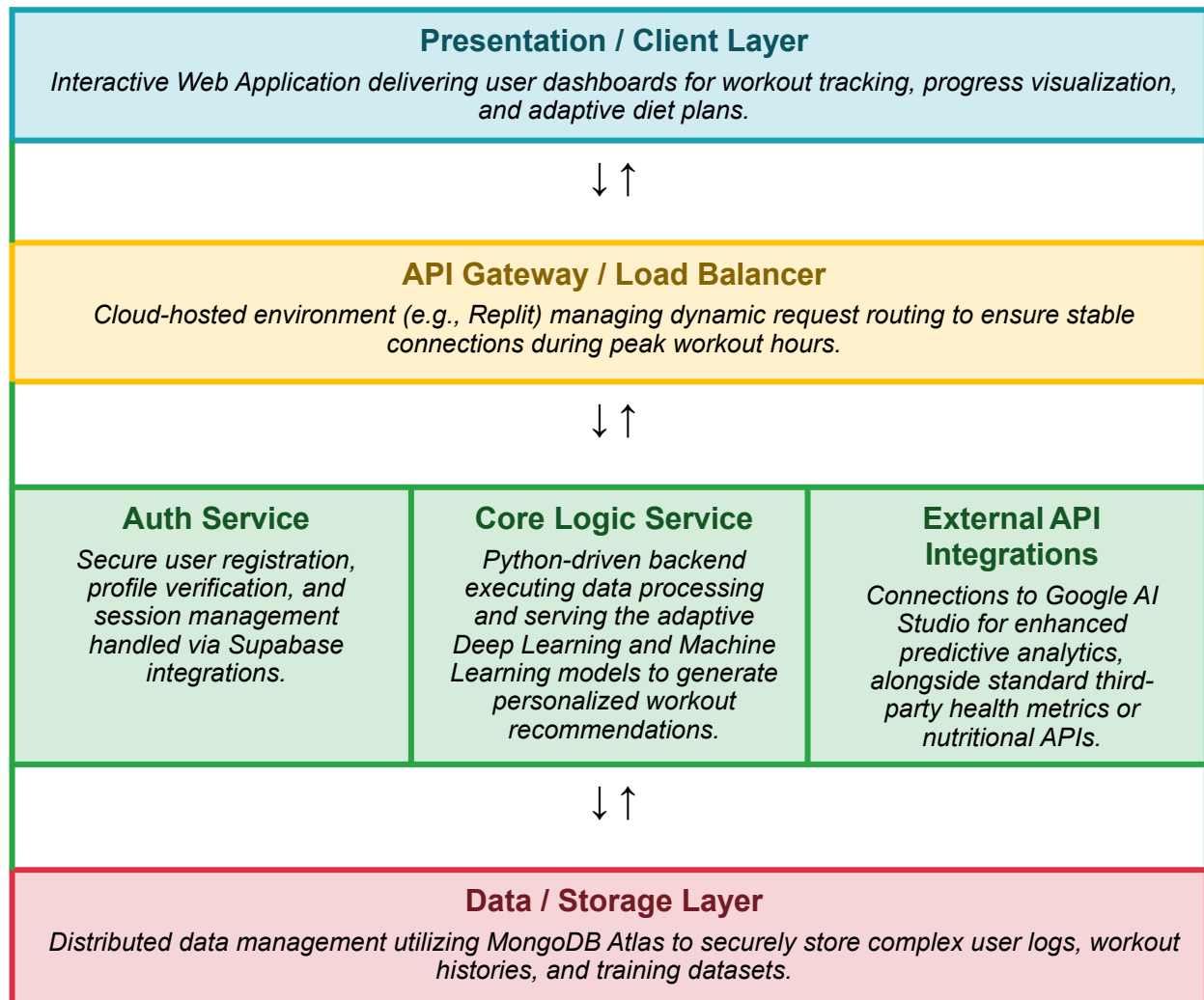


Solution Architecture

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-4644
Project Name	FIT BUDDY
Maximum Marks	5 Marks

Solution Architecture Diagram:

Here is the tailored architectural breakdown for the Fit Buddy application, mapped directly to the structural layout provided in the template.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides the interactive user interface for the fitness dashboard, allowing users to log workouts and view personalized training recommendations.</i>	<i>React, HTML/CSS, Replit (Hosting)</i>
<i>[API Gateway]</i>	<i>Serves as the primary entry point for client requests, securely routing traffic to the backend and ensuring system stability during peak traffic</i>	<i>Flask, REST APIs, Nginx</i>
<i>[Microservices]</i>	<i>Manages isolated backend processes, including secure user authentication and the execution of deep learning models for adaptive workout plans.</i>	<i>Python, Supabase, Google AI Studio, scikit-learn</i>
<i>[Database]</i>	<i>Securely stores and manages all persistent application data, including user profiles, session logs, and comprehensive training datasets.</i>	<i>MongoDB Atlas</i>